

As a final safety precaution, when a user goes to retrieve data, those copied shreds are compared to make sure they are all identical before the host gets paid for storage services. If something does get flagged, indicating that a node has been manipulated or the data has been altered, that user is immediately removed. There are many nodes required to make a quick transaction.

The numerous benefits of using blockchain in cloud computing are:

- Efficiency in ownership tracking
- Decentralisation with enhanced security
- Enhanced private key security and geo-independence
- Micro transactions and distributed supercomputers

Almost all cloud providers have launched blockchain cloud services to help organisations with blockchain projects. These include:

- Alibaba Cloud
- Google Cloud Platform
- IBM Cloud
- Microsoft Azure
- Oracle Cloud
- Amazon Web Services

Alibaba Cloud

Alibaba Cloud BaaS (Blockchain as a Service) is an enterprise-level Platform as a Service (PaaS) powered by the Ant Financial Blockchain Team, enabling business enterprises to build a stable, secure blockchain environment and support its management, operations, maintenance and development in an easy manner. It is built on Alibaba Cloud Container Service for Kubernetes clusters, and provides security, maintenance and computing. It provisions blockchain services based on multiple architectures such as public cloud deployments and private cloud deployments.

Alibaba Cloud BaaS supports open source blockchain technologies like Hyperledger Fabric and Enterprise Ethereum-Quorum. It also supports proprietary financial-grade blockchain

technology, Ant Blockchain. Its architecture is as follows.

- **Infrastructure layer:** Currently, BaaS supports public cloud and private cloud offerings of Alibaba Cloud. It will support hybrid cloud deployment in the near future.
- **Cloud resource layer:** It provides basic cloud resources for blockchain services and upper-layer applications, including ECS, VPC, NAS and SLB.
- **Platform services layer:** Built on Alibaba Cloud Container Service Kubernetes clusters, this blockchain platform supports multiple basic BaaS services. These services include resource creation, resource management, resource operation, and security management. The blockchain engines currently support Linux Foundation's Hyperledger Fabric 1.4 LTS, Ant Financial's Ant Blockchain, and J.P. Morgan's Enterprise Ethereum - Quorum.
- **Mid-layer application:** This is a reference architecture that is used to connect BaaS with business applications. It is usually implemented in the form of a blockchain business solution or blockchain middleware.

The overall architecture also includes multiple services that may be applicable to BaaS, such as security management and operation management.

Google Cloud

Google Cloud has collaborated with Digital Asset Holdings and BlockApps to support blockchain technology. It has also joined the private beta of Digital Asset's developer program to provide technology partners, software vendors, and financial organisations access to the software development kit (SDK) for DAML. The DAML (Digital Asset Modelling Language) is a smart contract language for distributed ledger technology (DLT) to align processes

across organisations. BlockApps Strato is based on the Ethereum protocol and provides enterprise-grade API integration capabilities and configurable consensus algorithms. It also enables enterprises to query and report on blockchain data using a traditional SQL database. Strato provides full integration with Google Cloud and rapid deployment for both cloud and hybrid infrastructure. This solution is developer-friendly, and enables developers to integrate it with existing enterprise systems.

IBM Blockchain Platform

The IBM Blockchain Platform for IBM Cloud is the next generation of IBM Blockchain Platform offerings, which gives users total control over their deployments, certificates, and private keys. It includes the new IBM Blockchain Platform console, which is a user interface that can simplify and accelerate the process of deploying components into a Kubernetes cluster on the IBM cloud, managed and controlled by the systems administrator.

Its key features are listed below.

- **Build:** It supports an integrated developer experience — deploys easy programming and DevOps as well as up-to-date Fabric key features.
- **Operate:** Total control on deployment — hosting or joining network, complete control on identities, unified operation, accessibility, easy interaction and log management, elimination of Docker, and Kubernetes service integration.
- **Grow:** Scalability, disaster recovery, and connectivity to other fabric networks.

Microsoft Azure

Azure Blockchain service is a fully managed ledger service that gives users the ability to scale and operate blockchain networks